

# Skills Summary

## Related Rates and Optimization: Getting the Setup Right

Use this reference while completing your worksheet or when studying.

### Skill 1 — Setting Up a Related-Rates Problem

**Order matters.** Write the geometric relation first, differentiate *with respect to  $t$* , and substitute the instant's numbers only at the end. Every variable that changes picks up its own rate:  $\frac{d}{dt}(r^n) = nr^{n-1}\frac{dr}{dt}$ .

Substituting  $r = 5$  before differentiating turns a variable into a constant and destroys the term you need.

**Example:** A circular ripple's radius grows at 2 cm/s. Find  $\frac{dA}{dt}$  when  $r = 6$  cm.

**Step 1.** Area and radius are linked by  $A = \pi r^2$ , and both change with time, so differentiate that relation with respect to  $t$  before any number goes in.

**Step 2.**  $\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(6)(2) = 24\pi$   
 $\Rightarrow \frac{dA}{dt} \approx 75.40$  square centimetres per second.

**Try it:** The same ripple grows at 3 cm/s. Find  $\frac{dA}{dt}$  when  $r = 10$  cm, to two decimals.

check: 188.50

### Skill 2 — Building an Optimization Model

**Two equations, then one.** Write the *constraint* (what is fixed) and the *objective* (what is optimized), use the constraint to remove one variable, then differentiate the objective and set the *derivative* — not the objective — to zero.

Check the geometry of the objective before differentiating: a wrong area or volume formula cannot be rescued by correct calculus.

**Example:** 100 m of fence encloses a rectangle on all four sides. Maximize the area.

**Step 1.** The perimeter is the constraint and the area is the objective, so solve the constraint for one side and put the area in a single variable.

**Step 2.**  $2x + 2y = 100$  gives  $y = 50 - x$ , so  $A(x) = x(50 - x)$  and  $A'(x) = 50 - 2x$ .  
 $\Rightarrow A'(x) = 50 - 2x$ , and  $A'(x) = 0$  at  $x = 25$  metres.

**Try it:** Same rectangle, 80 m of fence. Find  $A'(x)$  and the value of  $x$  that maximizes the area.

check:  $A'(x) = 40 - 2x$ ,  $x = 20$

## Skill 3 — Justifying That You Found the Extremum

**A critical value is not yet an answer.** Discard any solution outside the stated domain, then justify the survivor:  $f'' > 0$  means a minimum,  $f'' < 0$  a maximum, or use a sign chart for  $f'$ . On a closed interval, compare the critical values with both endpoints.

**Example:** Minimize  $f(x) = x + \frac{16}{x}$  on  $x > 0$ .

**Step 1.** Solving gives two critical values, so the real work is deciding which one the domain and the concavity allow.

**Step 2.**  $f'(x) = 1 - \frac{16}{x^2} = 0$  gives  $x = \pm 4$ ; only  $x = 4$  is in the domain, and  $f''(x) = \frac{32}{x^3} > 0$  there, so it is a minimum with  $f(4) = 4 + 4$ .

$\Rightarrow x = -4$  or  $4$ , minimum value  $f(4) = 8$ .

**Try it:** Minimize  $f(x) = x + \frac{25}{x}$  on  $x > 0$ : list both critical values, then give the minimum value.

check:  $x = -5$  or  $5$ ,  $f(5) = 10$